

- 3 Bears hibernate during the winter months, so are inactive for long periods of time.

The photograph shows two bears hibernating.



(Source: © Minden Pictures / Alamy Stock Photo)

Hibernating bears do not develop deep vein thrombosis (DVT).

Deep vein thrombosis is a blood clot that forms within a deep vein, often in the leg.

In humans, short-term periods of inactivity, such as sitting in an aeroplane, increase the risk of DVT.

This is due to reduced blood flow in the veins, causing damage to the walls of the veins.

- (a) (i) Which molecule, released from platelets, results in blood clotting?

(1)

- ☐ A fibrinogen
- ☐ B prothrombin
- ☐ C thrombin
- ☐ D thromboplastin

- (ii) Which pair of molecules are enzymes in their active form?

(1)

- ☐ A fibrinogen and prothrombin
- ☐ B prothrombin and thrombin
- ☐ C thrombin and thromboplastin
- ☐ D thromboplastin and prothrombin

(iii) Which molecule found in the blood is insoluble?

(1)

- ☐ **A** fibrin
- ☐ **B** fibrinogen
- ☐ **C** prothrombin
- ☐ **D** thromboplastin

(b) Pulmonary embolism can be caused by a DVT when the blood clot travels through the circulatory system and blocks the pulmonary artery.

Symptoms of pulmonary embolism include difficulty in breathing and a rapid heart rate.

Suggest why pulmonary embolism can result in these symptoms.

(3)

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- (c) A study of bears in Sweden discovered that these bears produced less HSP47 protein when hibernating in the winter.

This protein is found on platelets and is involved in the binding of platelets to damaged blood vessels.

Discuss whether this discovery could be useful to humans.

(3)

(Total for Question 3 = 9 marks)

